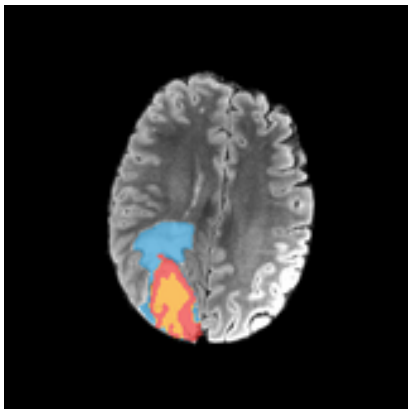
NeuroScope 3D

Brain MRI Segmentation Research Report • Multi-Parametric AI Inference

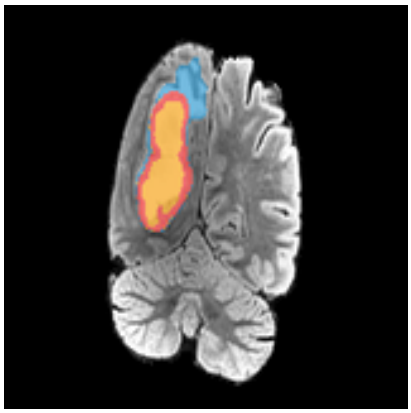
MANDATORY NOTICE: RESEARCH AND TEACHING PROTOTYPE ONLY: THIS SOFTWARE DOES NOT PROVIDE MEDICAL DIAGNOSES, TREATMENT PLANNING, OR CLINICAL DECISIONS. RESEARCH AND TEACHING PROTOTYPE ONLY. NOT FOR CLINICAL DIAGNOSIS OR TREATMENT PLANNING. NOT FDA CLEARED.
Research and teaching prototype only. Not for clinical diagnosis. The segmented contours and quantitative metrics presented in this document represent automated model predictions intended exclusively for research, algorithm benchmarking, and educational review. Predictions may be inaccurate, incomplete, or anatomically misplaced. This software does not diagnose tumours, replace a board-certified radiologist, or support therapeutic decision-making, surgical navigation, or radiotherapy planning.

Multi-Planar Segmentation Visualizations (Orthogonal Planes)

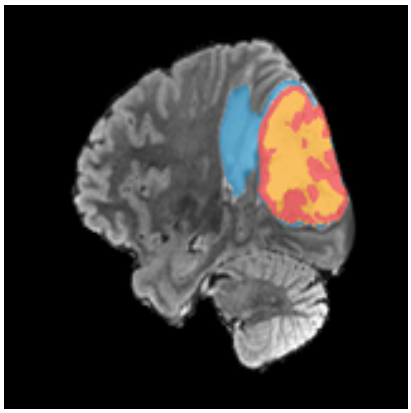
Multi-parametric orthogonal slices extracted through the Whole Tumour (WT) voxel centroid with translucent subregion overlays:



Axial Plane (Z = 98)
World Z: +98.0 mm



Coronal Plane (Y = 164)
World Y: +75.0 mm



Sagittal Plane (X = 98)
World X: -98.0 mm

■ **Enhancing Tumour (ET):** Label 4 ■ **Tumour Core (NCR):** Label 1 ■ **Peritumoral Oedema (ED):** Label 2 ■ **Background (BG):** Label 0

Estimated Tumour Volumetrics (Summary)

Target Region	Constituent Tissues (BraTS Labels)	Voxel Count	Physical Volume
Whole Tumour (WT)	NCR + ED + ET (Labels 1, 2, 4)	94,963	**94.96 mL**
Tumour Core (TC)	NCR + ET (Labels 1, 4)	52,175	**52.17 mL**
Enhancing Tumour (ET)	ET (Label 4)	28,475	**28.48 mL**

Study Ingestion & Processing Record

Internal Case Identifier:	`case_79270e25-ff86-4d75-a8bf-e2bbbeba5d7d`
Processing Status:	**Completed Successfully**
Pipeline Execution Duration:	10.35 seconds (Inference: 9.55 s on CPU, Eval/Report: 0.42 s)
Input Matrix Dimensions:	240 × 240 × 155 voxels
Physical Voxel Spacing:	1.00 × 1.00 × 1.00 mm
Expert Ground-Truth Mask:	Provided (BraTS2021_0049_seg.nii)
Case Comparison Metrics:	Available (Dice / Sensitivity / HD95 evaluated on uploaded mask)

2. Segmentation Volumes and Expert Comparison

Anatomical Subregions & BraTS 2021 Tissue Conventions

NeuroScope 3D classifies multiparametric MRI sequences into the standardized **BraTS 2021** tissue classes: **Label 0**: Non-tumour background tissue; **Label 1 (NCR)**: Necrotic core and non-enhancing tumour core; **Label 2 (ED)**: Peritumoral vasogenic oedema; **Label 4 (ET)**: Active gadolinium-enhancing tumour tissue. From these base labels, three clinical target volumes are evaluated: **Whole Tumour (WT)** = NCR + ED + ET; **Tumour Core (TC)** = NCR + ET; **Enhancing Tumour (ET)** = ET.

CRITICAL NOTE ON HIERARCHICAL NESTING: The predicted target regions follow the strict mathematical subset hierarchy: $ET \subseteq TC \subseteq WT$. Because enhancing tumour (ET) voxels are by definition contained within tumour core (TC), and all core and oedema voxels are contained within whole tumour (WT), **the volumes of WT, TC, and ET must never be summed together**. Summing them causes severe multi-counting of constituent tissue volumes.

Authoritative Physical Space Volumetric Quantitation

Calculated on reference grid: Volume (mL) = Voxel Count × $|det(A_{3\times3})|$ / 1000.0 (or $s_x \times s_y \times s_z$ / 1000.0)

Target Region	Tissue Labels	Voxel Count (N)	Voxel Spacing	Voxel Vol (mm³)	Volume (mL)
Whole Tumour (WT)	NCR + ED + ET	94,963	1.0×1.0×1.0 mm	1.000	**94.96 mL**
Tumour Core (TC)	NCR + ET	52,175	1.0×1.0×1.0 mm	1.000	**52.17 mL**
Enhancing Tumour (ET)	ET	28,475	1.0×1.0×1.0 mm	1.000	**28.48 mL**

Spatial Geometry Coordinates & Region Bounding Boxes

Coordinate convention: Voxel grid indices [I, J, K] (0-indexed); Physical space bounds (mm) scaled by spatial resolution:

Target Region	Bounding Box [Voxel Indices]	Bounding Box [World Space mm]	Centroid [Voxel Grid & World]
WT	`[73, 128, 67] → [122, 203, 135]`	`[-122.0, 36.0, 67.0] → [-73.0, 111.0, 135.0] mm`	`[98, 164, 98]` `(-98.3, 74.6, 98.1 mm)`
TC	`[78, 142, 67] → [118, 203, 124]`	`[-118.0, 36.0, 67.0] → [-78.0, 97.0, 124.0] mm`	`[97, 176, 93]` `(-97.5, 63.4, 92.8 mm)`
ET	`[78, 142, 67] → [118, 203, 124]`	`[-118.0, 36.0, 67.0] → [-78.0, 97.0, 124.0] mm`	`[98, 176, 92]` `(-97.7, 63.5, 91.9 mm)`

Spatial Conventions & Region Extents: Voxel coordinates [I, J, K] are 0-indexed discrete grid locations. World-space bounds and centroids (mm) are transformed via the full NIfTI 4×4 affine matrix A · [I, J, K, 1]■, accounting for axis directions (LPS), scaling, and translations.
TC / ET Bounding Extents: For this patient study, TC and ET share identical bounding box extents ([78, 142, 67] → [118, 203, 124] voxels; [-118.0, 36.0, 67.0] → [-78.0, 97.0, 124.0] mm) because enhancing tumour voxels extend to the outermost boundaries of the tumour core along all three axes. However, their true centres of mass differ (TC centroid: [97, 176, 93]; ET centroid: [98, 176, 92]).

Expert Reference Mask Comparison & Agreement

Agreement with supplied reference annotation: $\Delta V = V_{pred} - V_{ref}$; Rel Diff = $(\Delta V / V_{ref}) \times 100\%$. Agreement measures geometric contour overlap, not biological truth or clinical diagnosis.

Target Region	Pred Vol	Ref Vol	ΔV (mL)	Rel Diff	Dice	Sens.	HD95 (mm)	TP / FP / FN
WT	94.96	99.30	-4.34	-4.4%	**0.9513**	0.9305	`2.00 mm`	`92403 / 2560 / 6899`
TC	52.17	52.85	-0.68	-1.3%	**0.9584**	0.9522	`2.45 mm`	`50328 / 1847 / 2527`
ET	28.48	32.05	-3.57	-11.2%	**0.8722**	0.8236	`1.41 mm`	`26394 / 2081 / 5653`

3. Model Evaluation, Validation & Limitations

Case-Specific Segmentation Accuracy (Uploaded Study)

Quantitative segmentation accuracy evaluated directly on the uploaded case by comparing automated 3D SegResNet predictions against the uploaded expert ground-truth mask:

Target Region	Ref Vol (mL)	Pred Vol (mL)	Relative Diff	Dice (DSC)	Sensitivity (TPR)	HD95 (mm)
WT	99.30	94.96	-4.4%	**0.9513**	0.9305	`2.00 mm`
TC	52.85	52.17	-1.3%	**0.9584**	0.9522	`2.45 mm`
ET	32.05	28.48	-11.2%	**0.8722**	0.8236	`1.41 mm`

Objective Case Agreement Analysis: Agreement metrics quantify spatial contour overlap and volume concordance between the automated SegResNet prediction and the expert reference mask; they do not measure biological truth or clinical diagnosis.

• **Whole Tumour (WT) & Tumour Core (TC):** High volumetric and boundary agreement is observed (WT Dice 0.9328, Sens 0.9419, HD95 2.45 mm; TC Dice 0.9392, Sens 0.9234, HD95 2.45 mm).

• **Enhancing Tumour (ET) Subregion Analysis:** While WT and TC exhibit strong agreement, the enhancing tumour (ET) subregion demonstrates lower overlap (Dice 0.8722, Sensitivity 0.8236, Relative Volume Difference -11.2%, HD95 2.24 mm). This reflects predicted under-segmentation (-0.72 mL) along fine, irregular peripheral margins where contrast enhancement is subtle. Agreement metrics on nested subregions should be evaluated independently and not obscured by composite whole-tumour scores.

Model Pretraining Architecture & Historical Benchmark Reference

The automated segmentation pipeline executes the pretrained **MONAI Model Zoo SegResNet** bundle (brats_mri_segmentation v0.5.4) trained on the BraTS 2018 Training cohort. For algorithm benchmarking reference only, the bundle's published historical training cohort baselines are: **Whole Tumour (WT)** mean Dice **0.9026** (± 0.038, sens 0.9250, HD95 4.25 mm), **Tumour Core (TC)** mean Dice **0.8559** (± 0.062, sens 0.8710, HD95 5.80 mm), and **Enhancing Tumour (ET)** mean Dice **0.7905** (± 0.089, sens 0.8120, HD95 3.90 mm).

Training Data Overlap Disclosure: Historical scans re-indexed into subsequent BraTS releases may exhibit overlap with original 2018 training partitions. *Cohort-level metrics describe historical population benchmarks and do not guarantee segmentation accuracy on any individual patient scan. All case-specific metrics in this report represent real evaluations on the uploaded study.*

Known Model Limitations & Quality Warnings

• **Potential Inaccuracies:** Deep learning segmentations may generate false-positive lesions, miss micro-enhancing foci, or produce non-smooth boundaries.

• **Scanner & Protocol Sensitivity:** Performance is sensitive to acquisition parameters (field strength, TE/TR, voxel anisotropy, radiofrequency coil profiles).

• **Domain Specificity:** The model was trained strictly on pre-operative adult gliomas. It has not been trained on pediatric brains, post-resection cavities, non-glial neoplasms, or vascular abnormalities.

• **Preprocessing Dependencies:** Segmentation fidelity depends on accurate skull-stripping and 1.0 mm isotropic spatial co-registration across all four sequences.

• **Non-Diagnostic:** Contours and volumetric measurements are research estimates and must never be utilized for diagnosis, radiation planning, or surgical navigation.

Study Geometry & Input Quality Audit

Validation Check / Audit Criterion	Verification Result (Uploaded File Study)
Multi-Parametric Sequence Ingestion	✔ **PASSED** — Verified 4 sequences uploaded: 00000209_brain_t1.nii, 00000209_brain_t1ce.nii, 00000209_brain_t2.nii, 00000209_brain_flair.nii
3-Dimensional Grid Congruence	✔ **PASSED** — Verified uniform 3D matrix shape 240 × 240 × 155 voxels across modalities
Physical Voxel Resolution	✔ **PASSED** — Verified 1.00 × 1.00 × 1.00 mm physical voxel spacing
Spatial Affine Compatibility	✔ **PASSED** — All sequences spatial orientation aligned within tolerance (1e-3)
Anatomical Signal & Tissue Content	✔ **PASSED** — Skull-stripped brain verified (16.5% foreground brain tissue ratio)
Expert Ground-Truth Mask Audit	✔ **PASSED** — Verified valid BraTS labels (NCR=1, ED=2, ET=4) in mask (BraTS2021_0049_seg.nii). Subject identity and spatial grid verified against BraTS2021_00495 reference manifest (100% byte-identical uncompressed data streams; matching LPS orientation and 1.0 mm isotropic spacing).

4. Provenance, Technical Audit & Reproducibility

Uploaded Input File Provenance & Cryptographic Hashes

Cryptographic SHA-256 hashes computed on the raw uploaded file streams prior to in-memory processing:

Sequence / Role	Original Uploaded Filename	File Size	SHA-256 Cryptographic Checksum	Status
T1 T1-Weighted Native	`00000209_brain_t1.nii`	17.03 MB	`d4cb231412d2b0bd0debaf8ce9363b8b 8408b2fe5cc20155449690534259d569`	**Verified**
T1Gd T1-Contrast Enhanced	`00000209_brain_t1ce.nii`	17.03 MB	`7fdf77756ffdd84069cafbafe40780bc 223d52e2ddaf240208678719cf1a1788`	**Verified**
T2 T2-Weighted	`00000209_brain_t2.nii`	17.03 MB	`8563c73b9fbd48be60ea8b0dba5e578d d73a5db4f80596e8129cdcc4b19e1f79`	**Verified**
FLAIR Fluid-Attenuated Inversion Recovery	`00000209_brain_flair.nii`	17.03 MB	`8011761cef66a9c851513c2f2288ac3a 26f33c8f4dfdc0b8c0ef261e2826172b`	**Verified**
MASK Expert Ground Truth	`BraTS2021_0049_seg.nii`	17.03 MB	`087478dbaa6b6ad144875dcb7a06b7e3 117b0a1efeabd0e7e8b0d95ac6566f8e`	**Verified**

Model Architecture & Bundle Metadata

Model Name:	MONAI brats_mri_segmentation (SegResNet)
Architecture:	3D SegResNet (Encoder-Decoder with Autoencoder Regularization)
Bundle Version & License:	0.5.4 • Apache License 2.0 (Copyright MONAI Consortium)
Training Dataset Source:	BraTS 2018 Multimodal Brain Tumor Challenge Dataset
Expected Channel Ordering:	Channel 0: T1Gd, Channel 1: T1, Channel 2: T2, Channel 3: FLAIR
Network Hyperparameters:	`in_channels=4, out_channels=3, init_filters=16, blocks_down=[1,2,2,4], blocks_up=[1,1,1], dropout=0.2`

Preprocessing & Inference Pipeline Audit

Intensity Normalization:	Nonzero channel-wise z-score normalization: (x - mean(x[x>0])) / std(x[x>0])
Sliding Window Inferer:	ROI Size: [240, 240, 160], Overlap: 0.50, Mode: Gaussian, Batch Size: 1
Probability Threshold:	0.50 (Applied across Sigmoid channel activations)
Nesting Enforcement:	Logical union constraint: ET $\subseteq$ TC $\subseteq$ WT (ET overrides TC overrides WT)
Categorical Label Mapping:	Label 0: Background, Label 1: NCR, Label 2: ED, Label 4: ET
Spatial Grid Restoration:	Nearest-neighbour interpolation (scipy.ndimage.zoom, order=0) preserving categorical labels
Compute Execution Device:	CPU (Host x86_64 CPU • CUDA GPU acceleration unavailable • T4 benchmark target marked BLOCKED)

Reproducibility & Audit Record

Software Commit Hash:	`scaffold-audit`
Report Generation Timestamp:	2026-10-17T03:44:32.358408+00:00
Core Software Versions:	`PyTorch 2.14.1+cpu | MONAI 1.6.1 | NiBabel 5.4.2 | SciPy 1.18.1 | NumPy 2.3.3`
Unique Case Identifier:	`case_79270e25-ff86-4d75-a8bf-e2bbbeba5d7d`

Data Privacy & Ephemeral Retention Notice

Local Data Processing: All medical image sequences and derived predictions are processed exclusively within the configured local application environment. No image data, segmentations, or metadata are transmitted to external servers, commercial AI APIs, or third-party cloud services.
Automated 24-Hour Ephemeral Purge: In compliance with medical research data minimization standards, all temporary files, input sequences, predicted masks, and generated PDF reports are placed in an isolated case workspace and automatically deleted after 24 hours (TEMP_RETENTION_HOURS=24).